Topic 13 — Disaster Geography

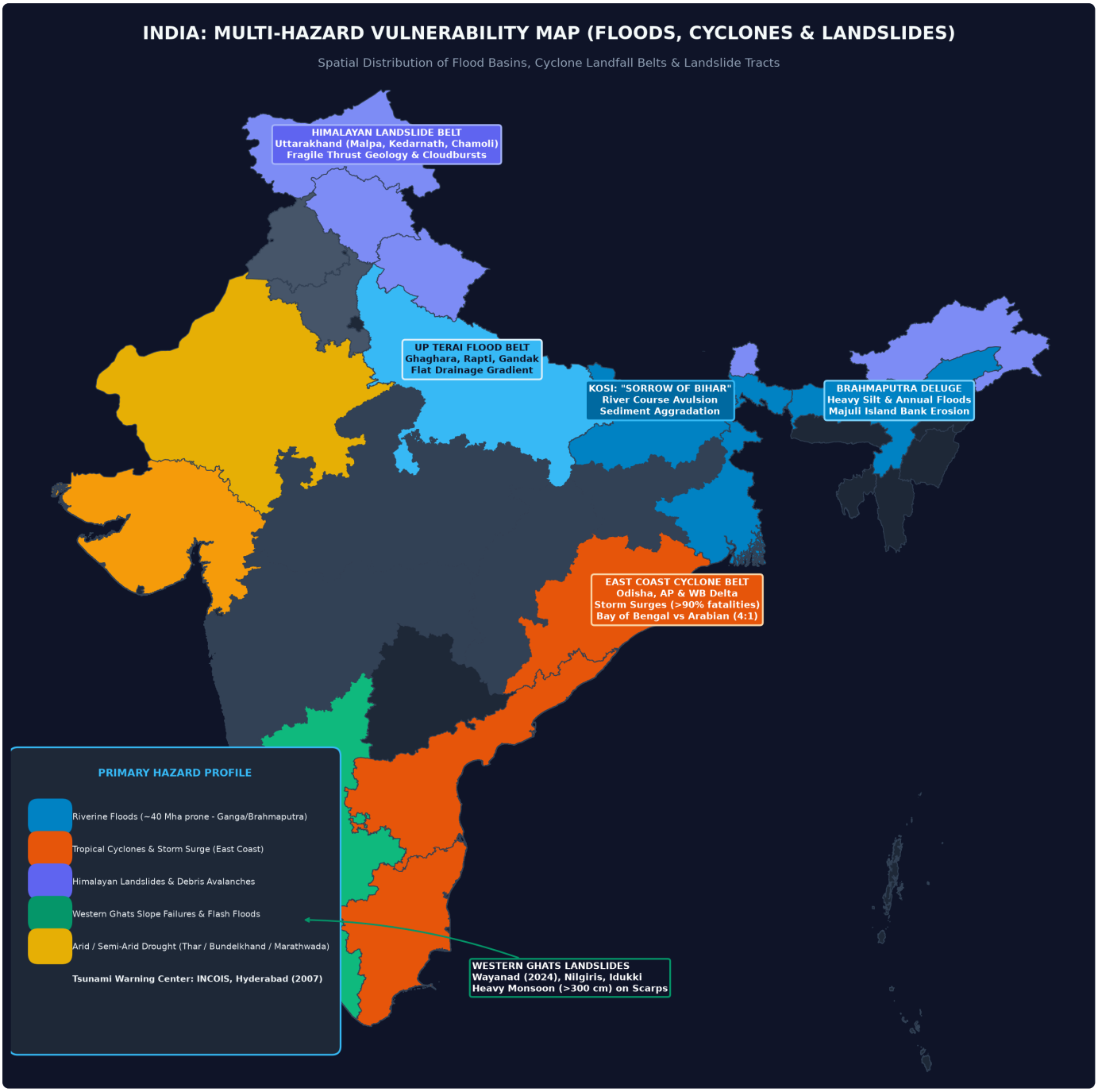
Sources baked in: NCERT (disasters / interior), PWOnlyIAS / Drishti DM notes, BIS IS 1893:2016, IMD, NDMA / DM Act 2005 / NPDM 2009, UPPCS Prelims 2018–2025 **Weight:** ★★★★ — seismic zones, NDMA chair, cyclone-name match, Himalaya A/R, IMD heat/cloudburst cuts; also teach full hazard map **Last verified:** August 2026 **Current Affairs:** BIS **Zone VI draft withdrawn Mar 2026** — map still **II–V (2016)**; Sendai 2015–30; CDRI 2019; Joshimath 2023; Sikkim GLOF 2023

Current Affairs (this topic)

Year	Fact	Why asked	Source
Mar 2026	BIS **IS 1893:2025 / Zone VI withdrawn**	Do **not** replace II–V in the hall	The Hindu / BIS
2015–30	Sendai Framework	Successor of Hyogo	UNDRR
2019	**CDRI** launched by India	Infrastructure + DRR	MEA / NDMA
2016 / 2019	NDMP issued / revised	First Sendai-aligned national plan	NDMA
2023	Joshimath subsidence; Sikkim GLOF	Landslide / flood CA	NDMA / news
2021	Tauktae (Arabian Sea → Gujarat)	“Arabian Sea never hits west coast” trap	IMD
2020	Amphan (BoB → WB/Odisha)	Super cyclone east coast	IMD

Cyclone **formation** (SST ~27°C, Coriolis, low shear) and IMD wind ladder are taught in N.10. Ring-of-Fire volcano list remains a Kilimanjaro trap pair.

N.0 Framework — Hazard, Vulnerability, Risk & DM Cycle



A **hazard** is a potentially damaging natural or man-made event. A **disaster** occurs when that hazard hits exposed people or assets and **capacity** is too weak to cope.

- Coaching fact: **Risk ≈ Hazard × Vulnerability / Capacity**.
- **Vulnerability** is how exposed and fragile a community, house or economy is.
- **Capacity** is the ability to prepare, resist, cope and recover.
- The same earthquake is a small event in a reinforced city and a disaster in a weak, crowded settlement.

Natural vs man-made

- Natural hazards include earthquakes, floods, cyclones, droughts, landslides, heat/cold waves, cloudbursts and tsunamis.

- Man-made / anthropogenic disasters include industrial and chemical accidents (**Bhopal 1984**), major fires, transport crashes and some CBRN risks.
- Climate change (जलवायु परिवर्तन) and unplanned urbanisation raise the chance that a hazard becomes a disaster.
- **Exposure** is who or what sits in harm's way; without exposure, a hazard may not become a disaster.

India's multi-hazard map (NDMA / NPDM note)

- About **59%** of the landmass is earthquake-prone (moderate (नरम दल) to very high).
- About **12%** of land is flood-prone.
- About **5,700 km** of the long coastline is cyclone / tsunami exposed.
- About **68%** of cultivable area is drought-vulnerable.
- Hills face landslides and avalanches; cities face urban floods.

Disaster management cycle

1. **Mitigation** — reduce risk before the event (codes, embankments, zoning, note).
2. **Preparedness** — plans, drills, early warning, stockpiles.
3. **Response** — rescue, relief, medical care during and just after impact.
4. **Recovery / reconstruction** — restore livelihoods and build back better.

India's institutional shift is from **relief-only** to **prevention, mitigation and Sendai-aligned risk reduction**.

N.1 Earthquakes — Mechanics

An earthquake is the sudden release of energy when rocks slip along a fault.

- The **focus (hypocentre)** is the rupture point **inside** the crust.
- The **epicentre** is the surface point directly above the focus.
- **Richter / moment magnitude** measures size or energy (logarithmic).
- The **Mercalli (MMI)** scale measures **intensity** — felt shaking and damage (Roman I–XII).
- Most damaging Indian quakes are **tectonic**. **Koyna 1967** is the classic **reservoir-induced** (RIS) trap.

About **90%** of world quakes sit on plate edges. The **Pacific Ring of Fire** is the circum-Pacific volcano–quake belt (Fuji, Pinatubo, Mount St. Helens). **Kilimanjaro is not** on that ring.

Seismic waves — body first, then surface

Seismic energy leaves the focus in two big families.

1. Body waves travel **through the interior** of the Earth. They split into **P** and **S**.

- **P-waves (primary / longitudinal)** are the **fastest**. They push and pull like sound waves and travel through **solids, liquids and gases**. They arrive **first** on a seismogram.
- **S-waves (secondary / transverse / shear)** are **slower** than P. They shake side-to-side and travel through **solids only**. They cannot cross the **liquid outer core**, so they arrive **second** where they are recorded.

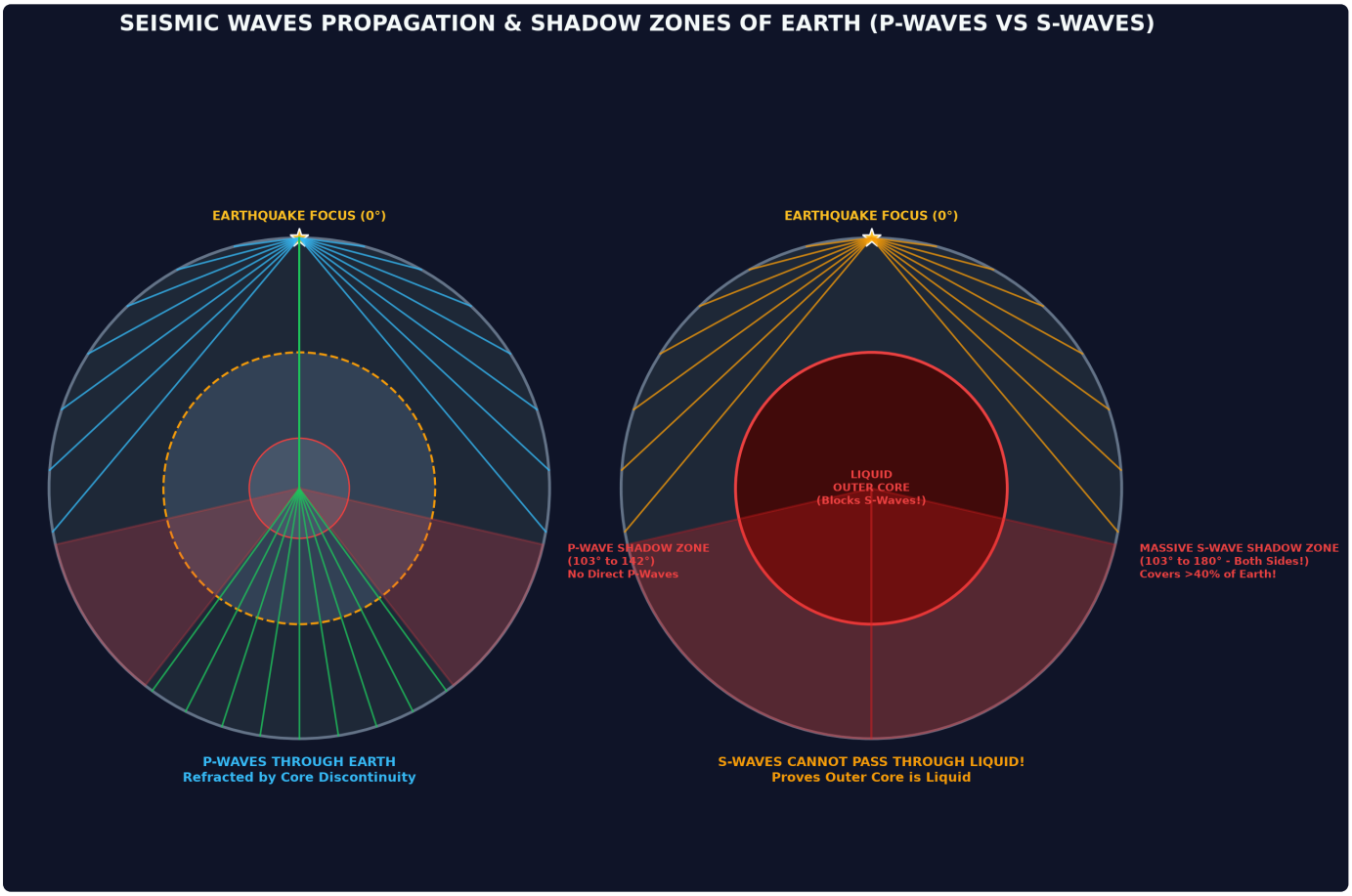
2. Surface waves travel along the **ground surface** after body waves.

- The two main types are **Love** and **Rayleigh** waves.
- They are the **slowest** to arrive but usually cause the **worst shaking and damage** at the surface.

Arrival order on a record is **P**, then **S**, then **surface** waves.

Wave	Family	Nature	Travels through	Arrival / damage
P (primary)	Body	Longitudinal / sound-like; **fastest**	Solids, liquids, gases	First
S (secondary)	Body	Transverse / shear; slower than P	**Solids only**	Second; stopped by liquid outer core
Love / Rayleigh	Surface	Along the free surface	Near-surface rock / soil	Last; usually most destructive at ground

Shadow zones



Each earthquake has its own shadow pattern because wave paths bend inside Earth.

- **S-wave shadow:** Beyond about **103–105°** from the epicentre, **S-waves are not recorded**. The liquid outer core blocks shear waves.
- **P-wave shadow: P-waves** are weak or absent in a belt of about **103–142°** (often taught as ~**105–140°**). They do enter the core but are **refracted** at the mantle–core boundary, so that angular belt is a gap.
- A claim that **both P and S are recorded at any distance beyond 105°** is **false**.

- The **focus** (or **hypocentre**) is the point inside the crust where the rupture begins.
- The **epicentre** is the point on the Earth's surface directly above the focus.
- **Richter** magnitude and **moment magnitude (Mw)** measure how much energy the quake released.
- The **Mercalli** scale measures **intensity** — how much shaking and damage people feel at a place.
- **Body waves** travel through Earth's interior; they are **P** and **S** waves.
- **Surface waves** travel along the ground; **Love** and **Rayleigh** waves are the main types.
- On a seismograph the arrival order is **P**, then **S**, then surface waves.
- The **S-wave shadow** begins beyond about **103°** from the epicentre because the liquid outer core blocks shear waves.
- The **P-wave shadow** is the gap of about **103–142°** where core refraction weakens or removes P arrivals.

Term	Recall
Focus / hypocentre	Inside the crust
Epicentre	Surface above the focus
Richter / Mw	Magnitude (energy)
Mercalli	Intensity (damage)
Body waves	Through Earth — **P** and **S**
Surface waves	Along ground — Love and Rayleigh
P then S then surface	Arrival order
S shadow	Beyond ~**103°** (liquid outer core)
P shadow	Gap ~**103–142°** (core refraction)

N.2 Earthquakes in India

The Himalayan arc is where the **Indian plate thrusts under Eurasia**, creating many **longitudinal thrust zones** (MCT, MBT, HFT).

- That is why the highest seismic concentration in India lies in the Himalaya.
- About **59%** of India's area is earthquake-prone.
- Peninsular India is **not** quake-free — **Latur 1993** and **Koyna 1967** are the hall traps.

Hall earthquakes — Cause, Course, Result

Koyna 1967 (Maharashtra)

Cause: Filling and operation of the **Koyna reservoir** loaded a stressed crustal block (classic **reservoir-induced seismicity**).

Course: A damaging shallow shock struck the Koyna belt in the Deccan.

Result: Proved peninsular India can shake hard. Do not answer “only Himalaya” for every Indian quake stem.

Latur / Killari 1993 (Maharashtra)

Cause: A hidden fault in the **stable Deccan** (Killari belt), not Himalayan collision.

Course: Shallow focus produced intense shaking across villages in Zone **III** country.

Result: The standard “peninsula is not aseismic” teaching card beside Koyna.

Bhuj 2001 (Gujarat)

Cause: Reactivation on the **Kutch** thrust / fault system in a high-hazard western pocket.

Course: A major shock on **26 January 2001** devastated Bhuj and surrounding towns.

Result: Reinforced **Kutch / Zone V** framing on the western map — different story from Latur’s “stable peninsula” trap.

Uttarkashi (उत्तरकाशी) 1991 (Uttarakhand)

Cause: Himalayan thrust seismicity in the Garhwal (गढ़वाल) arc.

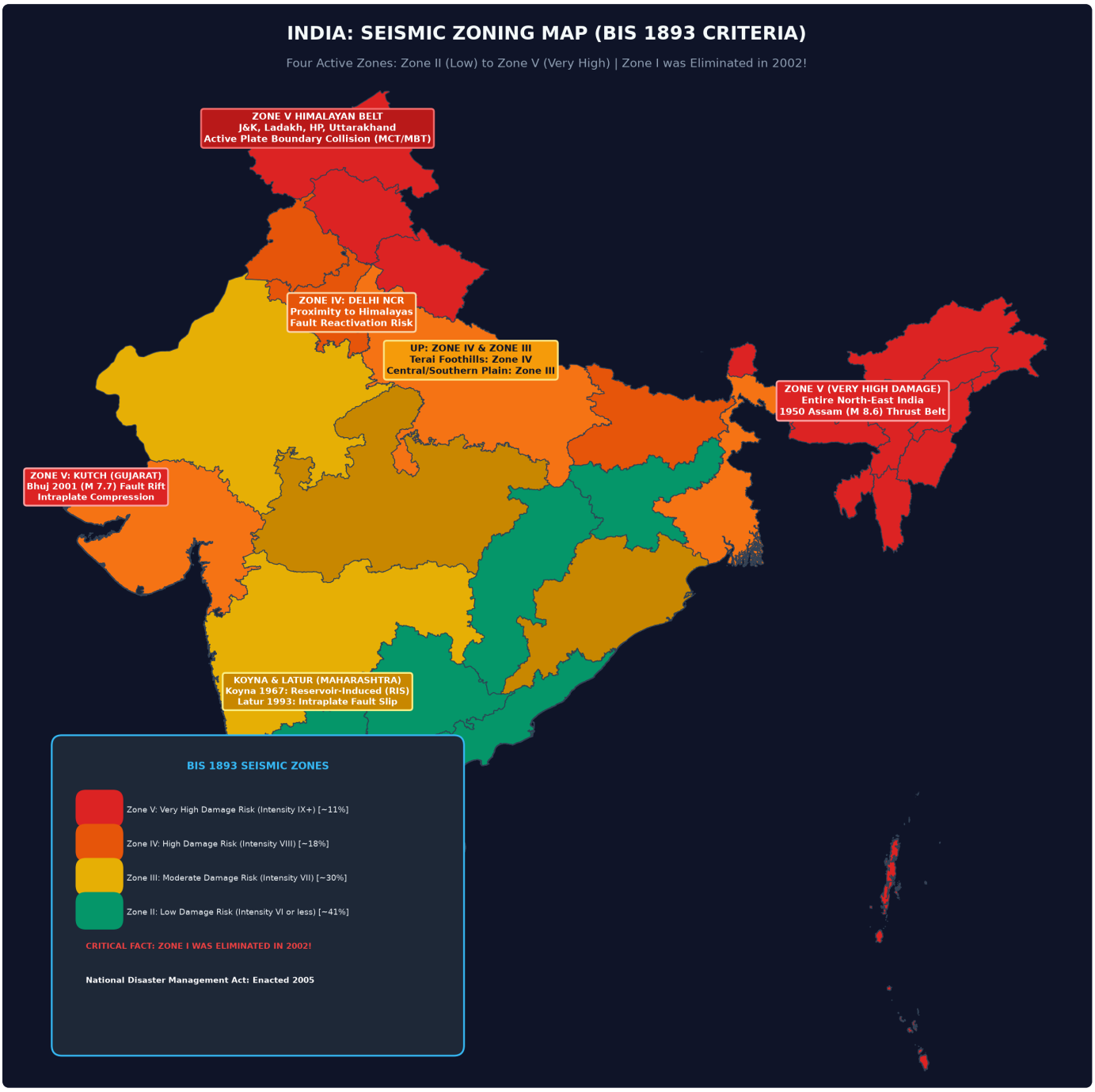
Course: A strong regional shock damaged hill towns and roads in Uttarkashi belt.

Result: A Garhwal Himalayan hall year, not a peninsular RIS card.

Year	Event	Trap
1819	Allah Bund / Kutch	
1897	Shillong / Assam	
1905	**Kangra**	
1934	**Bihar–Nepal** (east UP felt)	
1950	Assam–Tibet (~8.6)	
1967	**Koyna** (MH) — reservoir-induced	“Only Himalaya”
1991	Uttarkashi	
1993	**Latur / Killari** — stable peninsula	Himalayan
1999	Chamoli (चमोली)	
2001	**Bhuj** (26 Jan)	
2004	Sumatra–Andaman (tsunami)	
2005	Kashmir / Muzaffarabad	
2011	Sikkim	
2015	Nepal Gorkha (Bihar/east UP)	

> **Logic:** Peninsular India is **not** quake-free. Latur + Koyna are the two hall traps.

N.3 Seismic Zones of India



BIS **IS 1893 (Part 1): 2016** lists **four** (चातुर्याम) zones. **Zone I** was abolished in **2002**. On the present map **Zone V** is the **highest** hazard and **Zone II** is the **lowest**. A **Zone VI** draft (Nov 2025) was **withdrawn March 2026**. Prelims still use **II–V**.

Zone	Risk	Where (list)
V	Very high	NE India, Andaman & Nicobar, Himalayan pockets, Kutch, N Bihar

Zone	Risk	Where (list)
IV	High	Rest of Himalayan belt, Delhi, N/E UP Terai
III	Moderate	Broad peninsula, most of UP, many metros, Latur–Koyna
II	Low	Large stable Deccan / western Rajasthan interiors

- **Zone V** covers almost the entire **Northeast** (except some plains pockets on some maps), the **Andaman & Nicobar**, parts of **J&K / Ladakh (लद्दाख), Himachal and Uttarakhand**, the **Rann of Kutch**, and parts of north Bihar.
- **Zone IV** covers the rest of the Himalayan states, **Delhi**, northern and eastern **UP** (Terai / Ghaghara (घाघरा)–Gandak (गंडक) belt), and parts of Bihar, Haryana, Punjab, Maharashtra and J&K.
- **Zone III** covers the broad peninsula plus **most of UP** (Lucknow (लखनऊ), Kanpur (कानपुर), Prayagraj (प्रयागराज), Varanasi (वाराणसी), Agra). Mumbai, Chennai and Kolkata are often tagged III. The **Latur–Koyna** belt sits here too.
- **Zone II** covers large stable Deccan and western Rajasthan interiors (Hyderabad–Bengaluru belt).

UP: there is **no Zone V**. North/east Terai is **Zone IV**. The bulk of the state is **Zone III**.

> **Logic:** "Highest concentration in Himalaya" is true — that is Zone **IV–V**, not "all India Zone V."

N.4 Tsunami

A **tsunami** is a series of long ocean waves set off by **sudden seafloor displacement**. The word is Japanese: *tsu* (harbour) + *nami* (wave).

Causes

- Most tsunamis start from a large **undersea earthquake** on a **subduction / megathrust** fault that lifts or drops the seafloor.
- An underwater **landslide** can displace a huge water column and still raise a tsunami.
- A coastal or island **volcanic explosion / flank collapse** can also push water into a basin-crossing wave.
- Ordinary **wind waves** and cyclone swell are **not** tsunamis — they only disturb the surface layer.

How the wave travels

- In the open ocean a tsunami has a very **long wavelength** and low height, so ships (नावाध्यक्ष) may barely notice it.
- Speed is **high in deep water** (often hundreds of km/h) because the whole water column moves.
- Near the coast the wave **slows**, **shortens**, and **piles up** into a destructive surge or wall of water.
- The first sign on some shores is a sudden **drawdown** of the sea, followed by successive crests — not a single splash.

India — risk map

- Highest exposure sits on the **Andaman & Nicobar** arc, facing the Sumatra–Andaman subduction zone.
- The **east coast** (Tamil Nadu (नाडु), Andhra Pradesh, Odisha, West Bengal / Puducherry) and **Kerala** face open Indian Ocean wave paths.
- The **west coast** risk is lower but **not zero** — distant basin events and local undersea slips can still reach it.
- NDMA / multi-hazard notes flag about **5,700 km** of India's coastline as cyclone / tsunami exposed.

Early warning

- India's national tsunami warning hub is **INCOIS, Hyderabad** (Ministry of Earth Sciences).
- It uses seismographs, bottom-pressure sensors / buoys, and tide gauges to issue alerts to coastal states.
- After **2004**, India built this end-to-end system so that minutes of warning can drive evacuation.

26 December 2004 — Indian Ocean mega-event

Cause: A magnitude ~**9+** megathrust quake off **Sumatra** displaced the seafloor along the Sunda subduction zone.

Course: Waves raced across the Indian Ocean. **Andaman & Nicobar**, **Tamil Nadu**, **Puducherry**, **Andhra Pradesh** and **Kerala** were hit hard on the Indian side. Energy also reached Sri Lanka, Thailand, Indonesia and even Africa.

Result: Deaths of order **2–3 lakh** across the basin (not "50 lakh"). **Bangladesh** saw relatively few tsunami deaths because the main wave energy ran roughly **east–west** while that coast faces differently. India then hardened coastal warning and mangrove / shelter planning.

Item	Fact
Word origin	Japanese **harbour wave**
Trigger	Seafloor slip — quake / landslide / volcano
Not the same as	Wind waves / cyclone swell
Speed behaviour	Fast in deep water; piles up near coast
Warning hub	**INCOIS, Hyderabad** (MoES)
India high-risk coasts	A&N, east coast, Kerala; west coast lower but not zero
Hall event	**26 Dec 2004** — Sumatra megathrust
Soft shield	**Mangroves** / coastal forests cut surge and wave force

Mangroves and healthy coastal forests reduce cyclone storm-surge and tsunami impact by friction and by breaking wave energy before it reaches settlements.

> **Logic:** A tsunami is not a tidal wave. Tides follow the Moon and Sun; a tsunami follows sudden seafloor displacement.

N.5 Landslides

A **landslide** is the downslope movement of rock, debris or soil under gravity.

Why slopes fail

- Young, steep mountains with fractured rock fail more easily than old, gentle hills.
- Heavy rain or a **cloudburst** raises pore-water pressure and lubricates slip planes.
- Earthquakes shake loose material and can start rockfalls or debris flows.
- Road cutting, tunnelling, unplanned building and deforestation strip the root hold that binds soil.
- River undercutting at the toe of a slope removes support and invites collapse.

Types (coaching map)

Type	What moves
Fall	Rocks detach and freefall from a cliff
Slide	A mass moves on a distinct slip surface
Flow	Wet debris / mud moves like a fluid (**debris flow**)
Creep	Very slow downhill soil movement over years

Cloudbursts in the Himalaya often end as **debris flows** that bury roads and settlements.

High-risk belts in India

- The **Himalaya** (हिमालय) (Uttarakhand, Himachal, J&K / Ladakh, Sikkim, Arunachal) is the densest landslide belt.
- The **Western Ghats** (पश्चिमी घाट) fail after extreme monsoon rain (Kerala hill slides **2018**).
- The **North-Eastern hills** combine steep terrain with heavy rain.
- Soft sedimentary / weathered rock and seismic shaking raise risk further in the Himalayan arc.

Hall events — Cause, Course, Result

Malpa landslide (1998, Uttarakhand)

Cause: Steep fractured rock on the Kailash–Mansarovar pilgrimage corridor near **Malpa** (Pithoragarh (पिथौरागढ़) belt) failed after monsoon saturation and slope instability on a busy trek / road line.

Course: A major **rockfall / debris slide** buried the route and camps used by pilgrims and porters. Rescue was slow because the same slope kept shedding debris.

Result: Scores of travellers died. The event became a textbook Himalayan **rockfall-on-pilgrimage-route** card and pushed slope-safety talk on high Himalayan corridors.

Kedarnath disaster (June 2013, Uttarakhand)

Cause: Extreme monsoon rain and cloudburst-scale bursts over the **Mandakini** (मंदाकिनी) catchment raised pore pressure on already steep, glacier-fed slopes. Chorabari-type lake / debris release and slope failure fed the flood.

Course: Flash floods and **debris flows** tore through Kedarnath and the Mandakini valley. Roads, bridges, and settlements were buried or washed out across large stretches of Garhwal.

Result: One of India's worst Himalayan monsoon disasters. It reset state disaster planning, Char Dham corridor debate, and the national memory of cloudburst–landslide–flood cascades in Uttarakhand.

Kerala landslides and floods (2018)

Cause: Prolonged, extreme southwest monsoon rain soaked deeply weathered Western Ghats slopes. Cut slopes, quarrying, and settlement on steep belts raised exposure.

Course: Widespread **landslides** and debris flows ran with riverine flooding across Kerala's midland–highland districts. Reservoir releases and swollen rivers compounded inundation in the plains.

Result: Large loss of life and property. The year became the standard **Western Ghats monsoon-slide** teaching pair beside Himalayan cloudburst cards.

Joshimath subsidence (2022–2023, Uttarakhand)

Cause: Joshimath sits on old landslide / glacial debris with high water load. Construction, tunnelling and drainage stress on a fragile slope raised distress.

Course: Buildings cracked as the town showed slow **subsidence** and slope movement. Authorities evacuated pockets, halted some works, and mapped sink / crack zones.

Result: A living CA pair for **subsidence / slope distress**, not a single overnight rockfall. Do not merge Joshimath 2023 with Kedarnath 2013's rain–debris–flood cascade.

Mitigation note

- Avoid building on steep, freshly cut or historically sliding slopes.
- Keep drainage open so water does not pond inside the slope.
- Retain vegetation; use retaining walls, rock bolts and early-warning where needed.
- NDMA landslide guidelines stress mapping, land-use control and slope-safe roads.

> **Logic:** Joshimath **2023** is mainly **subsidence / slope distress**, while Kedarnath **2013** is a rain–debris–flood cascade. Do not treat every hill disaster as the same mechanism.

N.6 Floods

A **flood** occurs when water inundates land that is usually dry.

Types

- **Riverine floods** rise when large rivers overtop banks after prolonged monsoon rain or snowmelt. The **Ganga–Brahmaputra–Meghna** system dominates India's flood map.
- **Flash floods** rise in minutes to hours in hills after cloudbursts or sudden dam (बांध) / landslide lake bursts.
- **Urban floods** follow blocked drains, paved catchments and intense city rain (Mumbai-type events).
- **Coastal floods** come with cyclone **storm surge** and high tides.
- **GLOF** (glacial-lake outburst flood) releases a glacial lake when a moraine dam fails.

India riverine traps

- Watch **Kosi (कोसी), Gandak, Ghaghara and Rapti (राप्ती)** for east-UP / Bihar-style floods.
- The **Kosi** (कोसी) is famous for channel shift and embankment breaches (**Kosi 2008**).
- Brahmaputra and Barak (बराक) valleys flood widely in Assam and neighbouring hills.
- Eastern and trans-Ghaghara **Uttar Pradesh** (उत्तर प्रदेश) faces repeat monsoon inundation almost every year.

Kosi 2008 — Cause, Course, Result

Cause: Extreme monsoon rain plus stress on the embanked **Kosi** system near the Nepal–Bihar belt (**Kusaha** breach memory).

Course: The river jumped its embanked course and inundated vast stretches of **north Bihar**.

Result: Mass displacement and a lasting embankment-politics card. Pair it with Kerala 2018 as a Western Ghats monsoon flood–slide year, and Kedarnath 2013 as a Himalayan cloudburst cascade — three different mechanisms.

GLOF — living CA

- **Sikkim Teesta / South Lhonak 2023** is the standard glacial-lake outburst pair.
- Warming, moraine weakness and sudden drainage can send a wall of water down a narrow Himalayan valley.
- GLOF is a **flood** subtype, not a separate "non-flood" disaster class.

Institutions and hall years

- **CWC** (केंद्रीय जल आयोग) (Central Water Commission) and **IMD** run flood watch and rainfall alerts.
- State flood-control / irrigation departments manage embankments and relief coordination with SDMA / DDMA.
- Hall years: **Kosi 2008**, **Kerala 2018**, Kedarnath valley cascade **2013**, and east **UP** every strong monsoon.

Mitigation note

- Embankments, floodplain zoning, wetlands and early warning cut damage.
- Do not confuse flood **hazard** (water will come) with **disaster** (capacity and exposure fail).

N.7 Drought

A **drought** is a prolonged shortage of water relative (सापेक्ष) to what a place normally receives or needs.

Types

Type	Meaning
Meteorological	Rainfall well below normal
Hydrological	Rivers / tanks / groundwater fail
Agricultural	Soil moisture / crop stress
Socio-economic	Livelihood failure

- Meteorological drought (मौसमी (मौसम विज्ञान) सूखा) is the rainfall start of the chain.
- Hydrological drought (जल-वैज्ञानिक सूखा) follows when surface and groundwater stocks run down.
- Agricultural drought (कृषि सूखा) hits when soil moisture cannot support crops even if a tank still holds some water.
- Socio-economic drought (सामाजिक-आर्थिक सूखा) is when people lose work, food security or migration options.

IMD cut (meteorological)

- Seasonal rainfall **< 75% of normal** means deficiency **> 25%** — the usual meteorological drought gate.
- Moderate deficiency is about **26–50%**; severe is **> 50%**.
- Drought is declared for a season / region using these rainfall tests, not for one dry afternoon.

Belts

- Classic belts: **Bundelkhand**, Marathwada, Rayalaseema, Rajasthan, Kutch and Telangana interiors.
- In **Uttar Pradesh**, Bundelkhand and south-western pockets carry the drought label; the Terai is not a drought speciality.

Schemes chronology (asked order)

Scheme	Year
Community Development Programme	**1952**
DPAP (Drought Prone Areas Programme)	**1973–74**
Food for Work	**1977**
DDP (Desert Development Programme)	**1977–78**
TRYSEM	**1979**

Chronology used in **2025**: Community Development Programme (**1952**), then DPAP (**1973–74**), then Food for Work (**1977**), then TRYSEM (**1979**).

> **Logic: 1952** is CDP (सीडीपी), not DPAP. **DPAP** belongs to **1973–74**.

N.8 Cloudburst

A **cloudburst** is a sudden, very intense rain burst over a small area.

IMD definition

- Rainfall ≥ **100 mm in one hour** over a small area is the standard IMD cloudburst cut.
- Heavy monsoon rain over a wide plain is **not** automatically a cloudburst.

Where and why

- Typical belt: southern Himalayan rim at about **1000–2500 m**.
- Hotspots: **Uttarakhand (Garhwal)**, Himachal Pradesh, J&K, and North-Eastern hills.
- Windward **Western Ghats** can also see extreme short-period rain.
- Orographic lift forces moist air up steep slopes; unstable clouds dump rain in a narrow pocket.

Follow-on hazards

- Flash flood in a narrow valley.
- Debris flow and landslide on steep slopes.
- **Kedarnath 2013** is the classic rain–flash flood–debris cascade linked to extreme hill rainfall.

Mitigation note

- Early warning, slope-safe settlement and keeping nullahs clear reduce loss.
- Treat cloudburst as a **hill hazard**, not an Agra / Lucknow plains speciality.

> **Logic:** Heavy monsoon rain ≠ cloudburst. The cut is **100 mm/hour**.

N.9 Heat Waves & Cold Waves

Heat wave (IMD)

Declare a heat wave only if the **gate** temperature is crossed, then apply **departure** or **actual-T** cuts. Criteria use **two stations for two consecutive days**.

Region	Gate (Tmax)
Plains	≥ **40°C**
Coast	≥ **37°C**
Hills	≥ **30°C**

Region	Gate (Tmax)
Rule	Heat wave
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Departure from normal	**4.5–6.4°C**
Actual Tmax (plains)	≥ **45°C**

- Season: mainly **March–June** (rarely July).
- Heat often builds in north-west India and spreads east into the Gangetic plains.
- In Uttar Pradesh plains the hot dry wind is called the **Loo**.
- Cities raise risk further through the urban heat-island effect and outdoor labour exposure.

Cold wave (IMD)

- A cold wave is an abnormal drop in **minimum temperature** in winter (**December–February**).
- Plains are often gated near **Tmin ≤ 10°C**, then a sharp negative departure from normal.
- Severe cold wave applies when Tmin is very low or the negative departure is large.
- Western Uttar Pradesh and the Terai face cold plus dense **fog** in **December–January**.
- Fog cuts visibility, disrupts transport and raises health risk for the elderly and outdoor workers.

Confused pair

Pair	Correct
Heat wave vs any hot day	Need **gate** + departure / actual cut
Cold wave vs ordinary winter night	Need abnormal **Tmin** drop / departure
Loo	Hot dry plains wind of north India in summer

N.10 Cyclone Basins of India

A **tropical cyclone** is a warm-core low-pressure system over tropical seas with organised convection and strong winds.

Formation conditions (standard map)

- Sea-surface temperature about ≥ **26.5–27°C** over a deep warm layer.
- Enough **Coriolis force** — usually outside about **5°** of the equator.
- Low **vertical wind shear** so the storm tower can stay upright.
- Moist mid-troposphere and a pre-existing disturbance (trough / low).
- Mid-monsoon high shear often **suppresses** Bay of Bengal storms; peaks sit before and after the main monsoon.

IMD wind ladder (increasing intensity)

Stage	Approx. wind (km/h)
Low pressure area	< 31
Depression	31–49
Deep Depression	50–61
Cyclonic Storm	62–88
Severe Cyclonic Storm	89–117
Very Severe Cyclonic Storm	118–165
Extremely Severe Cyclonic Storm	166–220
Super Cyclonic Storm	≥ 221

Basins and seasons

Basin	Disaster fact
Bay of Bengal	**Higher frequency** (~4:1 vs Arabian in many decades); Odisha–AP–TN–WB–Sundarbans (सुंदरबन) surge
Arabian Sea	Fewer storms, but they **do** hit **Gujarat / Maharashtra**
Peaks	**May–Jun** and **Oct–Dec** (retreating monsoon (लौटता मानसून)). Mid-monsoon shear suppresses BoB

- Arabian Sea hits include **1998 Kandla**, **Tauktae 2021** and **Biparjoy 2023**.
- Storm surge, extreme rain and high winds cause most coastal deaths and damage.
- **Mangroves** and coastal forests cut surge and wind impact before water reaches settlements.

India cyclone events (teach the sequence)

- The **Odisha Super Cyclone of 1999** is the classic east-coast catastrophe for the state.
- **Phailin (2013)** struck Odisha as a very severe cyclonic storm with strong surge and rain.
- **Hudhud (2014)** made landfall near **Visakhapatnam** and damaged the Andhra coast.
- **Fani (2019)** hit Odisha as an extremely severe storm.
- **Amphan (2020)** was a Bay of Bengal super cyclone that hit **West Bengal and Odisha**.
- **Yaas (2021)** followed on the same eastern seaboard. Named-storm match practice sits in the bank below.

Local names (match bank)

Regional cyclone names already covered include Hurricane, Typhoon / Taifu, Baguio and Willy-willies. Also learn **Cyclone (North Indian Ocean)** and **Cordonazo**. ★ marks the high-yield match set.

Name	Country / basin	Note
Baguio / Baguios ★	**Philippines**	High-yield match
Hurricane ★	**USA** / Atlantic–Caribbean	High-yield match
Typhoon ★	NW Pacific — **China** class	High-yield match
Taifu ★	**Japan**-facing NW Pacific name	Often paired with Japan
Willy-willies ★	**Australia**	Cyclone ≠ Brickfielder (ब्रिकफील्डर)
Cyclone	North Indian Ocean (**India**/BoB/Arabian)	IMD names
Cordonazo	West Mexico Pacific	Coaching distractor

2019 named-storm match (core four) + nearby names

Storm	Most affected (fact)
Dorian ★	**Bahamas**
Hagibis ★	**Japan**
Lekima ★	**China**
Mitag ★	**Jeju, South Korea**
Amphan (2020)	West Bengal / Odisha
Fani (2019)	Odisha
Tauktae (2021)	Gujarat / west coast
Biparjoy (2023)	Gujarat / Arabian Sea
Odisha Super Cyclone (1999) / Phailin / Hudhud / Yaas	Classic India east-coast set

N.11 Disaster Management Framework

India has shifted from a **relief-only** habit to **prevention, mitigation and Sendai-aligned** risk reduction.

Legal and institutional spine

- The **Disaster Management Act, 2005** (notified **23 December 2005**) created the modern statutory architecture.
- The **National Disaster Management Authority (NDMA)** (राष्ट्रीय आपदा प्रबंधन प्राधिकरण) is the apex body; the **Prime Minister** is ex-officio Chairperson.
- The **National Policy on Disaster Management (NPDM)** came in **2009**.
- The **National Disaster Management Plan (NDMP)** was issued on **1 June 2016** and revised in **2019**, aligned to Sendai.

- The nodal Union ministry for disaster management is the **Ministry of Home Affairs** (Disaster Management Division).

Vertical structure

- **NDMA** sets national policy, guidelines and coordination. The Prime Minister chairs it. Membership is up to nine.
- **SDMA** — state apex; **Chief Minister** chairs.
- **DDMA** — district apex; **District Magistrate / Collector** chairs (elected district head often co-chairs).
- **NEC** (National Executive Committee) — **Union Home Secretary** coordinates implementation.
- **NDRF** — specialised response force under **MHA**, raised **2006** (began with 8 battalions; later expanded to **16**).
- **NIDM** — training and research from **New Delhi**. It grew out of the **National Centre for Disaster Management (NCDM)**, established in **1995** under the **Indian Institute of Public Administration (IIPA)** — not under ICSSR, and not as an NDMA body from day one.

Cycle (teach the order)

1. **Mitigation** — reduce risk before impact (codes, zoning, embankments, insurance, mangrove belts).
2. **Preparedness** — plans, drills, stockpiles, early warning.
3. **Response** — rescue, relief and medical care during and just after impact.
4. **Recovery / reconstruction** — restore livelihoods and **Build Back Better**.

International and CA pairs

- The **Sendai Framework (2015–2030)**, agreed at Sendai in Japan, follows the **Hyogo Framework (2005–2015)**. The earlier **Yokohama Strategy** dates to **1994**.
- It was adopted at the **Third UN World Conference on Disaster Risk Reduction**.
- Sendai sets **seven** global targets and **four** priorities for action. Do not accept “six targets and three priorities.”
- Sendai’s first priority is to **understand disaster risk**.
- The second is to **strengthen disaster risk governance**.
- The third is to **invest in disaster risk reduction** for resilience.
- The fourth is to **enhance preparedness** for effective response and to **Build Back Better** in recovery.
- **CDRI** (Coalition (गठबंधन) for Disaster Resilient Infrastructure) was launched by India in **2019**.
- International Day for Disaster Risk Reduction is **13 October**.

Inline PYQ — UPPCS Prelims 2021, Q26

Who is the Ex-officio Chairman of the National Disaster Management Authority?

A. The Prime Minister

B. The Home Minister

C. The Defence Minister

D. The Health and Family Welfare Minister

Show answer

Ans: A — NDMA is chaired by the **Prime Minister**.

Body	Role
DM Act	**23 Dec 2005**
NDMA	Apex body; the **PM** is ex-officio Chair; up to 9 members
SDMA	**CM** chairs
DDMA	District Magistrate chairs (elected head co-chairs)
NEC	National Executive Committee — **Union Home Secretary**
NDRF	**2006**, specialised response under **MHA**; started at 8 Bn and later rose to **16 Bn**
NIDM	Training / research in **New Delhi**; grew from **NCDM (1995, under IIPA)**
Nodal ministry	**MHA** (Disaster Management Division)
NDMP	**1 Jun 2016** (revised 2019) — aligned to Sendai
Sendai	**2015–2030**, Sendai (Japan), after **Hyogo 2005–2015** and Yokohama 1994; **7** global targets + **4** priorities; “Build Back Better” in Priority 4
NPDM	National Policy on Disaster Management — **2009**
CDRI	Coalition for Disaster Resilient Infrastructure — India, **2019**
IDDR	International Day for DRR — **13 October**
NCS	National Centre for Seismology — **MoES**
INCOIS	Tsunami — **Hyderabad**
IMD	Cyclone, rainfall, heat/cold

Sendai Framework — four priorities

1. Understanding disaster risk.
2. Strengthening disaster risk governance.
3. Investing in disaster risk reduction for resilience.
4. Enhancing disaster preparedness for effective response and to “Build Back Better” in recovery.

Industrial / chemical (man-made) disasters

- Man-made disasters include industrial explosions, toxic gas leaks, major fires, transport crashes and some CBRN risks.
- **Bhopal 1984** (Union Carbide MIC gas leak) is India’s classic industrial / chemical disaster pair.
- It is **anthropogenic**, not a natural hazard — a frequent multi-statement trap with cyclone / earthquake options.

- Prevention rests on zoning, process safety, on-site / off-site emergency plans and community awareness under DM and environmental rules.

Inline PYQ — UPPCS Prelims 2018, Q133

Which of the following statements is/are correct?

1. Developing countries suffer maximum damage from natural disasters.
2. The Bhopal gas tragedy was a man-made disaster.
3. India is a disaster-free country.
4. Mangroves help reduce the impact of cyclones.

A. 1 and 2 only

B. 2, 3 and 4 only

C. 1, 2 and 4 only

D. 1, 2, 3 and 4

Show answer

Ans: C — Statement 3 is false. India is **not** disaster-free. Bhopal is man-made; mangroves cut cyclone impact; developing countries often suffer higher damage from weak capacity.

Inline PYQ — UPPCS Prelims 2019, Q (Himalaya A/R)

Assertion (A): The highest concentration of seismic zones lies in the Himalayan region in India. Reason (R): There are many longitudinal thrust zones in the Himalayas.

A. Both A and R are true and R explains A

B. Both A and R are true but R does not explain A

C. A is true but R is false

D. A is false but R is true

Show answer

A/R logic: A tests India's highest seismic concentration in the Himalaya. R tests the many longitudinal thrust zones from Indian–Eurasian collision. R is the geological reason A is true. **Ans: A.** Both true, and R explains A — collision and thrust zones (MCT, MBT, HFT) concentrate Himalayan seismicity. This does **not** mean all of India is Zone V, and peninsular shocks like Latur still occur.

N.12 Uttar Pradesh Focus

Uttar Pradesh is a **multi-hazard** state: seismic shaking, river floods, Bundelkhand drought, plains heat, and winter fog — but **not** a whole-state Zone V or cloudburst plains label.

Seismic

- UP has **no Zone V** as a whole-state tag.
- North / east Terai and the Ghaghara–Gandak belt sit mainly in Zone **IV**.
- Lucknow, Kanpur, Prayagraj, Varanasi and Agra sit mainly in Zone **III**.

Floods

- Eastern and trans-Ghaghara districts face repeat monsoon floods.
- Watch **Ghaghara, Rapti, Gandak and Ganga** for riverine inundation.
- Embankments, drainage and early warning matter more here than Himalayan cloudburst maps.

Drought

- **Bundelkhand** and south-western / Vindhyan pockets carry the drought speciality.
- The always-wet Terai is **not** the drought belt.

Temperature extremes

- Plains **heat waves** and the **Loo** peak in **May–June**.
- Cold wave and dense fog peak in western UP and the Terai in **December–January**.

Cloudburst and hills

- Cloudburst is a **neighbouring Uttarakhand hill** hazard, not an Agra / Lucknow plains speciality.

Biological-disaster label (2018)

- In the 2018 UP paper, the **Gomati** carried the "biological disaster" pollution label among the given rivers — not Yamuna (यमुना) / Sai / Tamsa (तमसा) in that key.

Institutions

- **SDMA** under the **Chief Minister**; **DDMA** under the District Magistrate.
- State plans must knit flood, drought, heat and seismic guidelines into one preparedness map.

Hazard	Fact
Seismic	**No Zone V.** Terai / north-east is **IV**; Lucknow–Kanpur–Prayagraj–Varanasi–Agra belt is **III**
Flood	East UP — **Ghaghara, Rapti, Gandak, Ganga**; trans-Ghaghara belt
Drought	**Bundelkhand** + Vindhyan / south-west pockets
Heat	May–Jun plains (**Loo**)
Cold / fog	Western UP + Terai, Dec–Jan
Cloudburst	Not a plains speciality; hills of neighbouring Uttarakhand
Biological-disaster label (2018)	**Gomati** (pollution) — not Yamuna/Sai/Tamsa in that paper

Hazard	Fact
Framework	SDMA under **CM**; DDMA under DM